ZULHELMI SYAHTIAR

Fresh Graduate D3 Industrial Electronics

Cipayung, Jakarta Timur, Indonesia | 0838-9446-9333 | helmi2591@gmail.com
Portfolio | LinkedIn

SUMMARY

Fresh graduate in Industrial Electronics from Politeknik Negeri Jakarta with internship experience in power system protection at PT PLN. Strongly interested in the Internet of Things (IoT), microcontrollers, programming, and control systems. Skilled in hands-on microcontroller-based projects and simple automation system development. Eager to continuously learn, innovate, and apply modern technologies in industrial environments.

EXPERIENCE

PT PLN (Persero) ULTG Petukangan – South Tangerang *Jul 2025 – Oct 2025*
Protection Maintenance Intern (HAR PRO)

- Assisted with periodic maintenance and monitoring of substation protection systems.
- Performed testing of protection relays: OCR, GFR, Differential, Distance Relay, OVR/UVR, Buchholz, and Thermal Overload Relay.
- Involved in operational maintenance and system testing during DSS migration at a 150kV substation.
- Took part in Battery Capacity Tests (BCT) to maintain the reliability of substation backup power.
- Assisted with non-cascade testing on transformer incoming panels.

Independent Clients – Jabodetabek *Mar 2023 – Present*
Freelance Graphic Designer & Photographer

- Provided graduation and post-thesis-defense photography services, including style direction and photo editing.
- Created printed banner designs for clients' graduation celebration events.
- Created social media graphic designs for organizations' Instagram feed content.

Mediocreation – Brand Merchandise *Entrepreneurship Assignment*
Personal Project (Personal Brand)

- Designed the brand's visual identity, Instagram feed, and promotional posters.
- Created photo/video content for Instagram Story and Reels, and merchandise designs.
- Portfolio: @me.diocreation.

EDUCATION

Politeknik Negeri Jakarta – Depok, Jawa Barat *2023 – 2026*
D3 Industrial Electronics — GPA 3.55/4.00

- Final Project: "Dual DC Motor Control Trainer Kit – PID – Fuzzy Control Based on ESP32 with 3D Digital Twin Visualization".
- Sub-title: "Design of Web Monitoring and 3D Digital Twin Visualization Based on Real-Time Database for the Trainer Kit".
- Teaching Assistant for a Community Service project installing a solar-powered Hydroponic Green House.

SMAN 64 Jakarta – Jakarta Timur *2019 – 2022*
High School – Science Major (MIPA)

ORGANIZATIONAL EXPERIENCE

PSYCHOROBOTIC PNJ – Depok, Jawa Barat *Mar 2025 – Dec 2025*
Communication & Information Department
Student Study Group (KSM) at Politeknik Negeri Jakarta focused on Robotics and IoT.

- Created all design assets needed by the organization.
- Documented all organizational activities.
- Managed the organization's social media.
- Created educational content about robotics.
- Took part in robotics competitions.

E-TIME 2025 – Depok, Jawa Barat *May 2025 – Jul 2025*
Creative Media Division
E-TIME (Electro Activities Programme), a national-level electrical engineering competition held by the PNJ Electrical Engineering Student Association (Line Follower, PLC, Networking, Collaborative Robot).

- Created and prepared all design assets for the event: social media feeds, posters, merchandise, print designs, and certificates.
- Produced the event's aftermovie and other publication videos.

Olahraga Elektro 2024 – Depok, Jawa Barat *Apr 2024 – Aug 2024*

Publication and Documentation Division

Annual work program of HME PNJ; a yearly sports and arts competition for the PNJ Electrical Engineering Department.

- Documented all event activities.
- Managed the event's social media.
- Created and operated the photobooth at the closing ceremony.

EC-IKI PNJ 2024 Public Lecture – Depok, Jawa Barat *Nov 2024*

Media Division

Public lecture held by the Electrical Engineering Department for the D3 Industrial Electronics and D4 Industrial Control Instrumentation study programs.

- Documented all event activities.
- Ensured a smooth Zoom meeting experience for online participants.

FEATURED PROJECTS

- DC Motor Control Trainer Kit (2025–2026): Web monitoring dashboard (JavaScript ES6, HTML5, Tailwind CSS v4) and an interactive 3D Digital Twin (Three.js r160) synchronized in real time with ESP32 hardware via Firebase Realtime Database.
- Numa Space (2026): A virtual study room and real-time study-buddy matchmaking platform, with P2P video/audio interaction via WebRTC, Socket.IO chat, a private room access code system, and Pomodoro Timer integration.
- IoT Web Monitoring Dashboard (2025): Monitoring of temperature and soil & air humidity based on a MySQL database.
- Plant Monitoring Robot (2025): ESP32-C3 with DHT11 and soil moisture sensors, IoT-based.
- Object and Color Detection Mobile Robot (2026): Developed a real-time object and color detection system using a Raspberry Pi-based camera.
- Logic Gate Training Module (2024): Practice kit for AND, OR, NOT, NAND, NOR, XOR, Full Adder, Multiplexer, Demultiplexer, Flip-flop, and BCD-to-7-Segment Decoder.

CERTIFICATIONS

- **BNSP Competency Certification – Instrumentation Technician — BNSP (2026)**

Competent. Competency test: zero-span configuration and scaling of a level transmitter (4–20 mA) on a PLC, including linearity and measurement-deviation analysis.

- **TOEIC Prediction Test — Politeknik Negeri Jakarta (2026)**

Score 620/990 (Listening 260, Reading 360).

- **Creative Video Editing Techniques — Rumah Siap Kerja (Oct 2023)**

Credential No. A/1291/0383/PK002338/RSK/X/2023. Video editing workflow, transitions, and creative media content production.

- **Intensive Social Media Management Bootcamp — Growth Skill (Feb 2025)**

Credential No. 13/116/GROWTHSKILL/11/2025. Focused on personal branding, content strategy, graphic design, and social media management.

SKILLS

- **IoT & Microcontrollers:** ESP32, Arduino, Raspberry Pi, Firebase (Realtime Database, Firestore & Hosting), Computer Vision (OpenCV)
- **Control & Industrial Automation:** PID Controller, Fuzzy Logic Controller, PLC programming, SCADA (Schneider Electric, Vijeo Citect)
- **Protection & Testing Systems:** Protection Relays (OCR, Differential, REF MiCOM), OMICRON TestUniverse, GOOSE Configuration
- **Electronics Practical Skills:** Soldering, Desoldering, Wiring, Use of Measurement Tools (Multimeter, Oscilloscope, etc.)
- **Programming:** Python, C/C++, HTML, CSS, JavaScript, TypeScript, React, Node.js, Express
- **Design & Simulation:** NI Multisim, Proteus, AutoCAD, Fusion 360, Eagle
- **Visualization & Real-time:** Three.js, WebRTC, Socket.IO
- **Databases:** MySQL, PostgreSQL (Supabase)
- **Graphic Design, Multimedia & Photography:** Professional camera operation (directing & shooting), Canva, CapCut, Adobe Lightroom, Adobe Photoshop (basic), ibis Paint (basic)
- **Microsoft Office:** Word, Excel, PowerPoint
- **Languages:** Indonesian (native), Malay, English (TOEIC Prediction 620/990 – working proficiency)